

Reading the Front Page: Evaluating Multimodal LLM Grounding on Italian Newspapers

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Abstract

While multimodal large language models (LLMs) can now analyze document images and answer user queries, it is often unclear whether their responses are grounded in the actual visual evidence on the page. We evaluated two models, Claude Sonnet 5 and GPT-5.5, using 30 questions per newspaper across the front pages of ten Italian newspapers. The questions covered metadata, layout, text comprehension, and image analysis. For each task, the models were required to provide an answer, pinpoint the visual evidence used, and explain their reasoning. We evaluated these outputs combining a manual qualitative analysis of the models’ responses with automatic quantitative metrics based on textual similarity, such as ROUGE and BERTScore (via an Italian BERT model). The findings suggest that modern vision-language models often provide correct answers and reasonable explanations when evaluated by humans. However, automatic similarity metrics, while faster to apply, reveal a larger discrepancy between answer accuracy and evidence matching. This is especially true for complex document layouts, where a model may provide correct evidence using different words from the human annotations. These results suggest that current multimodal LLMs have good grounding capabilities, but additional evaluation methods, such as LLM-as-a-judge, could help better measure their ability to use visual evidence.

1 Introduction

This report presents the research conducted during a 4-week internship at the Natural Language Processing (NLP) Unit of Fondazione Bruno Kessler (FBK). The internship focused on evaluating the reasoning and explainability capabilities of state-of-the-art multimodal Large Language Models (LLMs) on complex newspaper front pages.

Multimodal LLMs can now directly process document images and answer questions without requiring optical character recognition (OCR) or spe-

cialized layout-parsing pipelines. This capability is highly valuable for tasks such as digitizing historical archives, monitoring media coverage, and conducting journalistic research, domains where documents feature dense text interwoven with complex layouts and varied imagery.

However, generating a correct answer does not guarantee that a model actually “read” the document. A model might guess the answer based on typical newspaper structures or fabricate a plausible explanation retroactively. This discrepancy between *answering correctly* and *accurately citing the source* is a critical gap that most multimodal LLM evaluations overlook, as they primarily focus on final answer accuracy.

To investigate this gap, we analyzed the front pages of ten Italian newspapers. Newspaper front pages serve as an excellent stress test because they densely consolidate diverse visual elements, including small print (edition numbers, prices, editors’ names), large headlines, standard paragraphs, and multimedia graphics.

We designed 30 questions per newspaper, categorized into four tasks: (1) Metadata and Identification, (2) Structure and Layout, (3) Reading and Comprehension, and (4) Images and Graphics. Models were prompted to provide an *answer*, the *evidence* (where the information appeared on the page), and an *explanation* linking the two. Outputs from Claude Sonnet 5 and GPT-5.5 were then scored against human-annotated ground truth.

Results showed that GPT-5.5 generally outperformed Claude Sonnet 5, but both models struggled with exact evidence overlap. Furthermore, performance dropped on one specific newspaper, indicating that complex page layouts and accurate evidence citation remain open challenges in multimodal document understanding, even when raw answer correctness is high.

2 Contribution During the Internship

During the internship, I contributed to the design and implementation of an evaluation framework for multimodal Large Language Models on newspaper front pages. My activities included:

- manually annotating newspaper evidence and compiling reference answers for the gold standard;
- developing and running experiments with GPT-5.5 and Claude Sonnet 5 through their respective APIs;
- implementing automatic evaluation metrics (e.g. ROUGE) to analyze the experimental results;
- developing the project code in Python using tools such as VS Code and Google Colab;
- using GitHub for distribution of the implemented software, including scripts, prompts and evaluation tools;
- interpreting the experimental findings and writing this report in LaTeX using Overleaf.

3 Methods

3.1 Data and Question Design

Our dataset comprises the front pages of ten Italian newspapers published on the same date, chosen to span the Italian political spectrum from left-leaning to right-leaning outlets and to include both general-interest and specialized (financial) publications: *La Repubblica*, *Corriere della Sera*, *La Stampa*, *Il Sole 24 Ore*, *Il Messaggero*, *Il Mattino*, *Il Giornale*, *Liberi*, *La Verità*, and *Il Fatto Quotidiano*. For each newspaper we constructed a spreadsheet and a corresponding JSON template containing 30 questions, divided into four task categories:

- **Task 1 – Metadata and Identification (7 questions):** Extracts the masthead name, publication date, edition number, price, and editor-in-chief.
- **Task 2 – Structure and Layout (11 questions):** Identifies the largest headline, article counts, photo placements, and the spatial proportions of page elements.
- **Task 3 – Text and Comprehension (5 questions):** Involves named-entity recognition within main headlines, thematic summarization, and verbatim transcription of opening sentences.
- **Task 4 – Images and Graphics (7 questions):** Requires counting photos, verifying captions, and describing charts or other visual data.

Questions were deliberately mixed between closed formats (bare-letter multiple choice, numeric counts) and open formats (verbatim transcription, free-text description). Closed-format questions constrain the answer space and make lexical scoring more reliable; open-format questions, particularly verbatim transcription, were included specifically to force the model to commit to a concrete, checkable string rather than a generic paraphrase, which is what makes the evidence field a meaningful probe of grounding rather than of general reading comprehension.

3.2 Model Querying

I queried Claude Sonnet 5 and GPT-5.5 through their respective provider APIs using a shared codebase of per-provider scripts (`claude_analyze_image.py`, `gpt_analyze_image.py`) that each take a prompt, a blank JSON question template, and a newspaper front-page image as input. The images were obtained by converting the newspaper PDF files into JPGs at 300 DPI resolution via <https://www.ilovepdf.com/>, using the “High” export setting. The scripts return a completed JSON file

in the same schema as the template. The codebase also includes PDF-input variants for Claude and GPT (`claude_analyze_pdf.py`, `gpt_analyze_pdf.py`); the PDF variants were exploratory and applied to only one newspaper, since the study’s focus was image-based reading.

To optimize context and API cost, I batched all questions belonging to a single task into one call together with the newspaper’s front-page image, resulting in four API calls per newspaper per model (one per task) rather than one call per question. This kept each call focused on a coherent sub-task while still allowing the model to see the full page image at every call.

Both models received an identical Italian-language system prompt instructing them to: rely strictly on visible image information and bypass external prior knowledge; output “N/A” whenever the requested information was missing or illegible; adhere to strict formatting constraints (e.g., bare letters for multiple-choice answers, verbatim text for transcription questions); and return a structured JSON object containing an `AI_Answer`, `AI_Evidence`, and `AI_Explanation` field for every question. Enforcing a fixed JSON schema made it possible to parse and score every model response automatically, without manual post-processing of free-form outputs.

3.3 Reference Annotation

Human annotators, including the author of this report, reviewed every front-page image and answered the complete question set, following annotation guidelines that mirrored the constraints given to the models (verbatim transcription where applicable, “N/A” for missing or illegible information). For each question the annotators recorded both the correct answer and the exact visual evidence span supporting it. These annotations served as the ground truth against which both models were evaluated.

3.4 Evaluation Metrics

To construct a comprehensive picture of model capabilities, our evaluation framework combined two complementary methodologies:

1. **An automatic quantitative evaluation** applied separately to the generated *Answer* and *Evidence* fields, measuring how closely they aligned with human reference annotations.
2. **A manual qualitative evaluation** where human annotators evaluated the *Answer* along with the combined *Evidence* and *Explanation* fields. This manual review (recorded as a unified score in the spreadsheet’s `Score` column) verified the conceptual accuracy, logical coherence, and visual grounding quality of the models’ reasoning pathways.

For the automatic quantitative pipeline, two distinct text-similarity metrics were calculated via the `evaluate.py` script:

- **ROUGE**, computed with stemming, capturing lexical (n-gram and longest-common-subsequence) overlap with the reference; we report the ROUGE-L F-measure, though ROUGE-1 and ROUGE-2 were also computed and stored for completeness.
- **BERTScore**, computing token-level similarity using contextual embeddings from an Italian BERT model (`dbmdz/bert-base-italian-xxl-cased`), averaged across all 12 layers without baseline rescaling, capturing semantic similarity that survives valid paraphrasing; we report the BERTScore F1.

Two rules handled missing values consistently across both metrics: if both the model and the reference field were “N/A,” the pair was scored as a perfect match (1.0); if only one of the two was empty, the pair was scored as a complete mismatch (0.0). We also defined, but have not yet populated, a third evaluation tier: an LLM-as-judge score intended to capture semantic correctness independent of lexical overlap. This tier is included in the scoring schema for forward compatibility and will be explored in future work.

The evaluation script writes per-question and aggregated summary scores directly back into the source spreadsheet alongside a detailed JSON metric registry, so that every score reported in Section 4 can be traced back to the specific model output and reference annotation it was computed from.

3.5 Reproducibility

All prompts, JSON question templates, model-querying scripts, per-newspaper model outputs, and the evaluation script are released in a public repository [1], together with a worked example (front-page image, filled JSON template, and the exact script invocation used to produce it) for a single newspaper.

4 Results

4.1 Overall Answer, Evidence, and Manual Overlap

Table 1 details the ROUGE-L and BERTScore metrics averaged across all 300 questions, alongside the average human qualitative scores. GPT-5.5 consistently outperformed Claude Sonnet 5 across both the automated and manual metrics.

Crucially, while both models exhibited a severe performance drop when transitioning from the automated *Answer* field to the automated *Evidence* field (dropping by approximately 0.3 in ROUGE-L and 0.25–0.27 in BERTScore), their human-evaluated manual scores paint a much more robust picture. In the manual qualitative evaluation, both models achieved high scores of **0.86** and **0.79** respectively. This indicates that despite superficial variations in exact word choices, the models systematically provided correct answers supported by logically sound and grounded explanations.

Model	Answer		Evidence		Manual Score
	ROUGE-L	BERTSc.	ROUGE-L	BERTSc.	
GPT-5.5	0.769	0.864	0.464	0.601	0.86
Claude Sonnet 5	0.718	0.844	0.413	0.573	0.79

Table 1: Quantitative and qualitative overlap with human reference, averaged across all ten newspapers.

4.2 Scores by Task

As shown in Table 2, model strengths varied by task. GPT-5.5 peaked on Task 2 (Structure & Layout), whereas Claude Sonnet 5 performed best on Task 3 (Text & Comprehension). Both models struggled significantly with Task 4 (Images & Graphics), which typically requires open-ended, descriptive reasoning rather than concise factual extraction.

Task	GPT-5.5	Claude Sonnet 5
1. Metadata & Identification	0.749	0.693
2. Structure & Layout	0.829	0.739
3. Text & Comprehension	0.792	0.822
4. Images & Graphics	0.657	0.589

Table 2: ROUGE-L F-measure (Answer field) broken down by task and model.

4.3 Variation Across Newspapers

Performance fluctuated significantly depending on the publication. Both models achieved their highest scores on *La Stampa* (GPT-5.5: 0.888; Claude Sonnet 5: 0.790 ROUGE-L). Conversely, both models performed poorest on *Il Mattino* (GPT-5.5: 0.558; Claude Sonnet 5: 0.575 ROUGE-L)—dropping roughly 0.2 to 0.3 points below their averages. This isolated drop suggests that specific typographic or layout complexities in *Il Mattino* hindered extraction, rather than a generalized model limitation.

Question	Human (Gold)	GPT-5.5	Claude Sonnet 5
D1.1 Qual è il nome completo della testata?	Answer: La Stampa Evidence: “La Stampa” riportato a caratteri cubitali al centro, in alto.	Answer: La Stampa Evidence: “La Stampa” riportato nella testata principale del giornale.	Answer: La Stampa Evidence: Il nome della testata è riportato nella parte superiore della prima pagina.
D1.1 Qual è il nome completo della testata?	Answer: IL MATTINO Evidence: “IL MATTINO” riportato a caratteri cubitali al centro, in alto.	Answer: IL MATTINO DEL LUNEDÌ Evidence: IL MATTINO DEL LUNEDÌ	Answer: IL MATTINO DEL LUNEDI Evidence: IL MATTINO DEL LUNEDI

Table 3: Representative qualitative comparison between the highest-performing newspaper (*La Stampa*) and the lowest-performing newspaper (*Il Mattino*). While both models correctly identify the newspaper title in *La Stampa*, they systematically include the subtitle “DEL LUNEDÌ” for *Il Mattino*, reducing lexical overlap.

5 Discussion

5.1 The Grounding Gap vs. Metric Sensitivities

The most prominent finding of this study is the severe disparity between Answer and Evidence scores across both GPT-5.5 and Claude Sonnet 5 when using automated evaluation. While the models demonstrate high accuracy in producing the correct final response, their ability to extract the precise visual evidence string annotated by humans appears suboptimal.

However, incorporating the manual qualitative scores reveals that this apparent "grounding gap" is heavily driven by metric sensitivities. Standard word-overlap metrics like ROUGE-L penalize any stylistic variation or paraphrasing of the evidence block, even if the model successfully located and cited the correct visual sector of the page. Because BERTScore (which captures semantic meaning) scored the evidence fields higher than ROUGE-L (0.601 vs. 0.464 for GPT-5.5), and because the manual qualitative evaluations remained consistently high, we conclude that the models’ actual visual grounding capabilities are much stronger than automated text metrics indicate. True grounding failures (such as retroactive justification or visual hallucination) do occur, but harmless semantic phrasing differences account for a substantial portion of the automated scoring drop. Implementing the planned “LLM-as-judge” evaluation tier will be crucial to automatically decoupling semantic correctness from strict lexical matching.

5.2 Layout Vulnerability and Typographic Stress-Testing

The sharp performance decline on *Il Mattino* (where GPT-5.5 and Claude dropped to 0.558 and 0.575 ROUGE-L, respectively) reveals that layout complexity is a major hurdle that general vision-language benchmarks fail to isolate. As demonstrated in Table 3, both models struggled with basic metadata extraction for this specific newspaper, systematically confusing the primary masthead (*IL MATTINO*) with the day-specific subtitle printed directly below it (*DEL LUNEDÌ*).

In contrast, the clean, centralized layout of *La Stampa* presented no such obstacles, allowing both models to achieve near-perfect alignment. This disparity highlights a visual processing flaw: current multimodal architectures struggle with hierarchical, non-standard typographic structures. When text blocks are closely integrated or overlapping, the models lose the ability to distinguish primary headers from secondary sub-headings. This suggests that document processing models cannot rely solely on raw OCR capabilities; they require a deeper, spatially-aware understanding of layout architecture to navigate the complex design systems typical of traditional print journalism.

5.3 The Bottleneck of Open-Ended Visual Reasoning

The results from Task 4 (Images and Graphics) further expose the structural limitations of modern vision-language models. This task, which yielded the lowest overall ROUGE-L scores (0.657 for GPT-5.5 and 0.589 for Claude Sonnet 5), required models to count photos, verify captions, and describe graphical charts.

Unlike closed-format questions in Task 1, Task 4 demands an integration of spatial mapping, visual counting, and open-ended synthesis. The low scores are indicative of two overlapping challenges:

- **High-density spatial counting:** Document images often feature multiple small, embedded photos or visual assets. The visual encoders of Claude Sonnet 5 and GPT-5.5 struggle to accurately count these small, scattered objects, often double-counting prominent visuals or completely skipping minor inline graphics.
- **Descriptive misalignment:** When describing a complex graphic or photo, the model’s free-text output naturally diverges from the precise, objective descriptions provided by human annotators. Because there are many valid ways to verbally describe an image, standard string-similarity metrics penalize these responses harshly, even when they are conceptually accurate.

These limitations highlight that open-ended visual reasoning remains a major bottleneck. Future work must focus on developing evaluation paradigms that can assess visual comprehension without being overly sensitive to specific phrasing.

6 Conclusion

This study evaluated Claude Sonnet 5 and GPT-5.5 on a rigorous 30-question benchmark covering metadata, layout, text comprehension, and images across the front pages of ten Italian newspapers. GPT-5.5 achieved the highest overall performance on both answer generation and evidence extraction. However, both models exhibited a substantial drop in accuracy when required to identify the exact visual evidence supporting their answers.

These findings suggest that, although modern multimodal LLMs can often produce correct answers, reliably grounding those answers in the source document remains a significant challenge. The shared failures observed on specific publications such as *Il Mattino* further indicate that complex typography and page layout present difficulties beyond simple text recognition. Future work should complement lexical similarity metrics with LLM-as-judge evaluations to better distinguish acceptable paraphrases from genuine grounding errors, and extend this evaluation framework to other dense, visually complex document types beyond Italian newspaper front pages.

Beyond the research activities, the internship provided an opportunity to gain practical experience with artificial intelligence tools, software development workflows, and the research environment at Fondazione Bruno Kessler. The student also had the opportunity to visit FBK facilities, including the clean room where semiconductor technologies are developed, gaining a broader view of the different research areas carried out within the institution.

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References

- [1] Samuel Fey. Github Repository with all necessary information to reproduce the experiments. <https://github.com/S-fey/Can-AI-read-newspapers>.